IAL EVALUTION STINT-2

By Ankit Yadav and Aditya Chauhan

AI-Powered Internal Operations

Two enterprise automation projects purpose eliminating manual bottlenecks in crew operations at IndiGo Airlines.

The Manual Problem We Set Out to Solve

Two high-friction workflows were consuming critical hours of crew operations staff time — with error-prone, non-auditable outcomes.

Crew Relocation Routing

Manual flight booking, AIMS code application, and hotel logistics across a 3-day relocation window — tracked in spreadsheets with no automated validation.

URTI Leave Processing

URTI requests handled via email, requiring manual balance checks, roster verification, and leave rule interpretation — all without a consistent audit trail.

Crew Rostering Assignment Workflow

Our fully automated Python and FastAPI workflow transforms the manual processing of crew roster allocations into a streamlined, efficient, and auditable process. From email input to AIMS-ready output, this solution handles complex logic and edge cases, dramatically reducing processing time from hours to minutes.

1

Email Parsing & Data Extraction

Unstructured `.msg` files containing crew batch data are automatically parsed using advanced regex patterns.

2

Assignment Verification

Base assignments are cross-referenced and verified against the AIMS API (MCP) to ensure data integrity.

3

Intelligent Flight Routing

Flights are routed via the JOC API, applying intelligent scheduling rules, prioritizing morning departures (7-10 AM for returns).

4

AIMS Code Application

Relevant AIMS codes (RLL, RLLH, DHT, OFG, N/A) are automatically applied based on predefined logic.

5

AIMS-Ready CSV Export

A compliant CSV file is generated, ready for AIMS integration, with built-in human confirmation checkpoints for oversight.

This solution effectively manages complex scenarios like dummy flights for own-transport and provides backup flight selections, ensuring operational resilience.

URTI: Email-Triggered Agentic Workflow

Our automated leave balance update process, powered by Copilot Studio agents, ensures efficient and accurate handling of URTI requests.

1

Email Trigger & Intake

Crew emails for example first.last@goindigo.in with "URTI REQUEST". Power Automate instantly detects the new email and initiates the workflow.

2

Email Parsing & Validation

Copilot Studio extracts key details (IGA ID, Name, Dates)automatically rejecting invalid requests.

3

Data Validation & Query

The Validation Agent queries the CLMS Database (via MCP Server Layer) to fetch crew info, detect calendar buckets, check URTI balances, and verify roster status.

4

Intelligent Decision Routing

The system routes requests based on scenarios: normal flow, bucket reversal, LWP escalation, or insufficient balance. Post-write verification confirms accuracy.

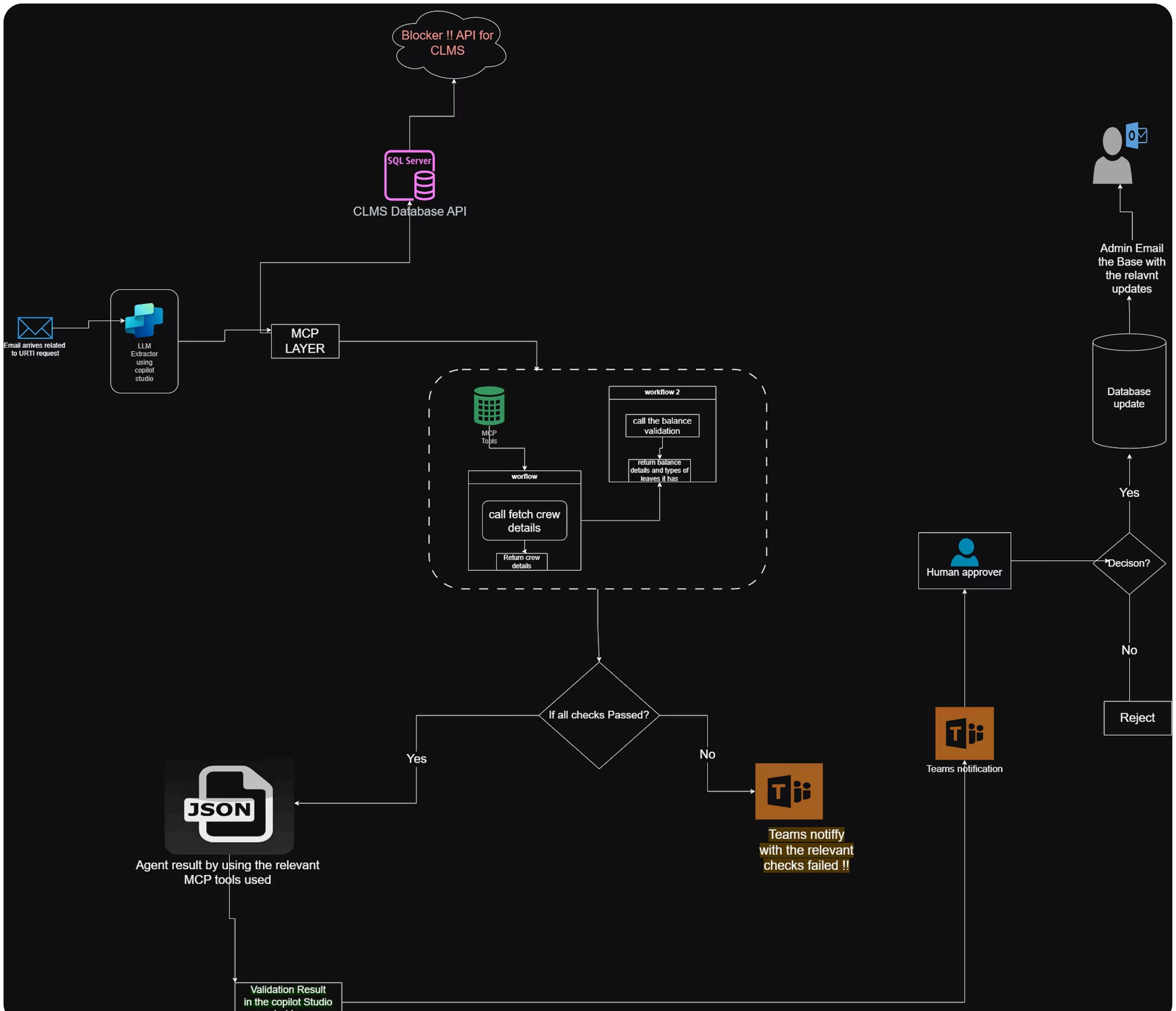
5

Notification & Audit

Crew receives confirmation Notification

Architecture is Shared in next page

Architecture For Solution Diagram



Crew Relocation Pipeline: The 3-Day Process

From iFly final batch to active duty — every step automated, AIMS codes applied correctly, and edge cases handled without manual intervention.

1 — Day 1 — RLL + DHT

Crew departs iFly. IFS routes via AIMS, applies RLL code. DHT applied at boarding — critical for location tracking. 1 primary + 2 backup flights booked.

2 — Day 2 — OFG

Off-day at hometown or base. OFG code applied in AIMS. No IFS action needed.

3 — Day 3 — RLLH

Flight routed hometown to base (7–10 AM window). RLLH code applied — mandatory for hotel allocation. Logistics books 13 nights.

4 — Days 4–5 — N/A

N/A code applied for license/AEP compliance. Crew cleared for active duty.

Key Results & Impact

Both projects will deliver measurable operational improvements(if **not considering the blocker of the resources?**) and are aligned with IndiGo's compliance and standards.

01

Full Process

Full relocation routing end-to-end has been understand and implemented using the python and fastapi services.

02

AIMS Codes

RLL, RLLH, DHT, OFG, N/A — applied using the pipeline with all rules intact as it is.

03

Automation

Full URTI Process can be automate if the issue of the MCP,
under the copilot studio agent is solved.

Key Learnings & Strategic Takeaways

Agentic AI Works in Production

Copilot Studio + Power Automate handles complex, rule-driven decisions reliably —with rollback built in from day one.

Edge Cases Are the Real Product

Own-transport crews, dummy flights, past-dated requests — the automation value is in handling what humans misses.

AI workflows understanding

Manual Workflow can be rewritten and been automate with the help of AI if done logically with taken care of all business requirements.

Scalable Architecture for More Use Cases

FastAPI microservices + Python pipelines provide a reusable pattern for future crew ops, IFS automation projects.

THANK YOU